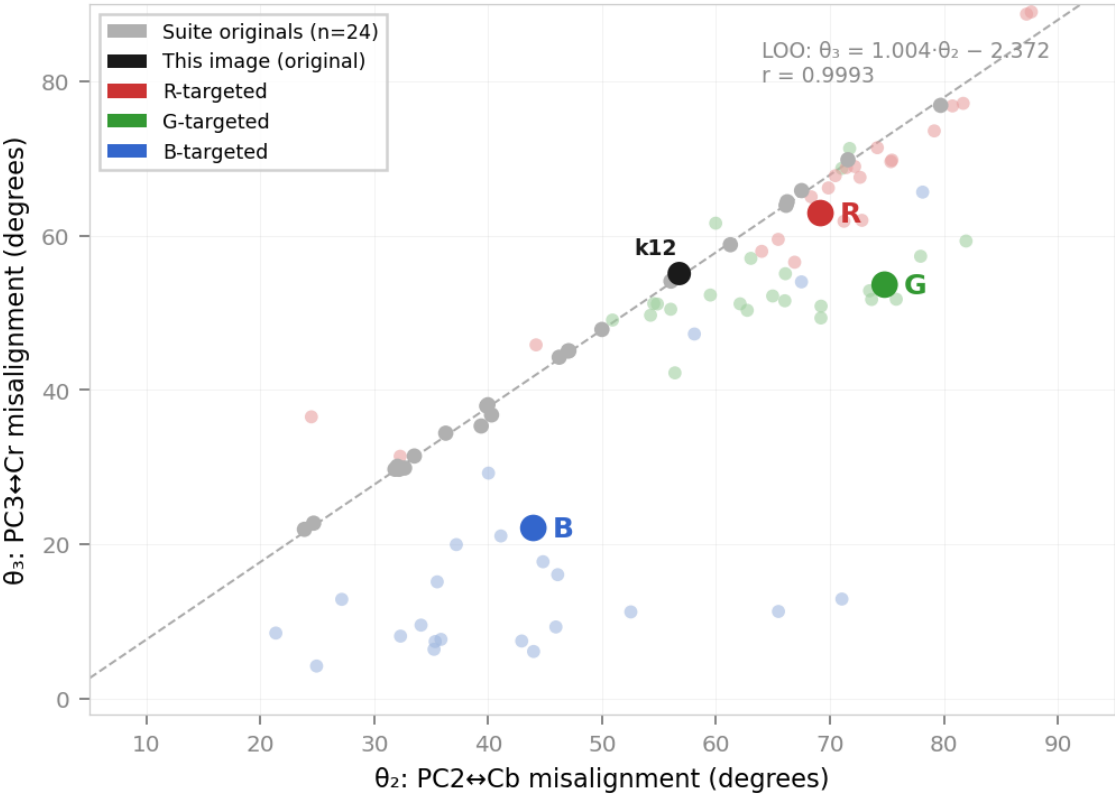


Compression Profile: KODIM12 768x512 | 24-bit RGB | PNG (lossless)

couple on beach | Strongly One-Dimensional (PC1 = 95.09%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.9133	0.8721	0.9674	0.918	56.77°	55.11°	+0.48°
R-targeted	0.1301	0.0607	0.9536	0.382	69.17°	62.95°	-4.12°
G-targeted	-0.0692	0.9793	-0.0121	0.354	74.8°	53.67°	-19.06°
B-targeted	0.9202	-0.1691	-0.0967	0.395	43.96°	22.12°	-19.64°

R-targeted isolates red (R-G, R-B < 0.13) while preserving G-B coupling at 0.954.
G-targeted isolates green (R-G, G-B < 0.07) while preserving R-B at 0.979.
B-targeted preserves R-G at 0.920 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold (r = 0.9993).

Compression Profile: KODIM12 768x512 | 24-bit RGB | PNG (lossless)

couple on beach | Strongly One-Dimensional (PC1 = 95.09%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

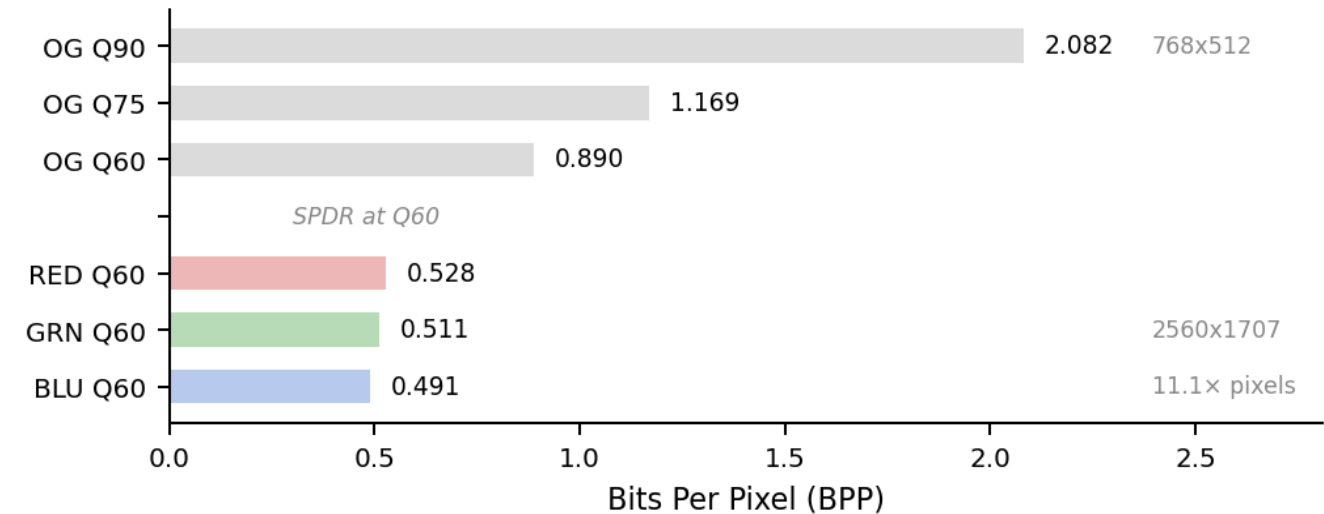
ORIGINAL (kodim12)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	768x512	518.6 KB	10.804	393,216
JPEG Q90	JPEG	768x512	99.9 KB	2.082	393,216
JPEG Q75	JPEG	768x512	56.1 KB	1.169	393,216
JPEG Q60	JPEG	768x512	42.7 KB	0.890	393,216

REDISTRIBUTED (kodim12 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	2560x1707	281.8 KB	0.528	4,369,920
GRN Q60	JPEG	2560x1707	272.7 KB	0.511	4,369,920
BLU Q60	JPEG	2560x1707	262.1 KB	0.491	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1x the original pixel count.

Compression Profile: KODIM12 768x512 | 24-bit RGB | PNG (lossless)

couple on beach | Strongly One-Dimensional (PC1 = 95.09%)

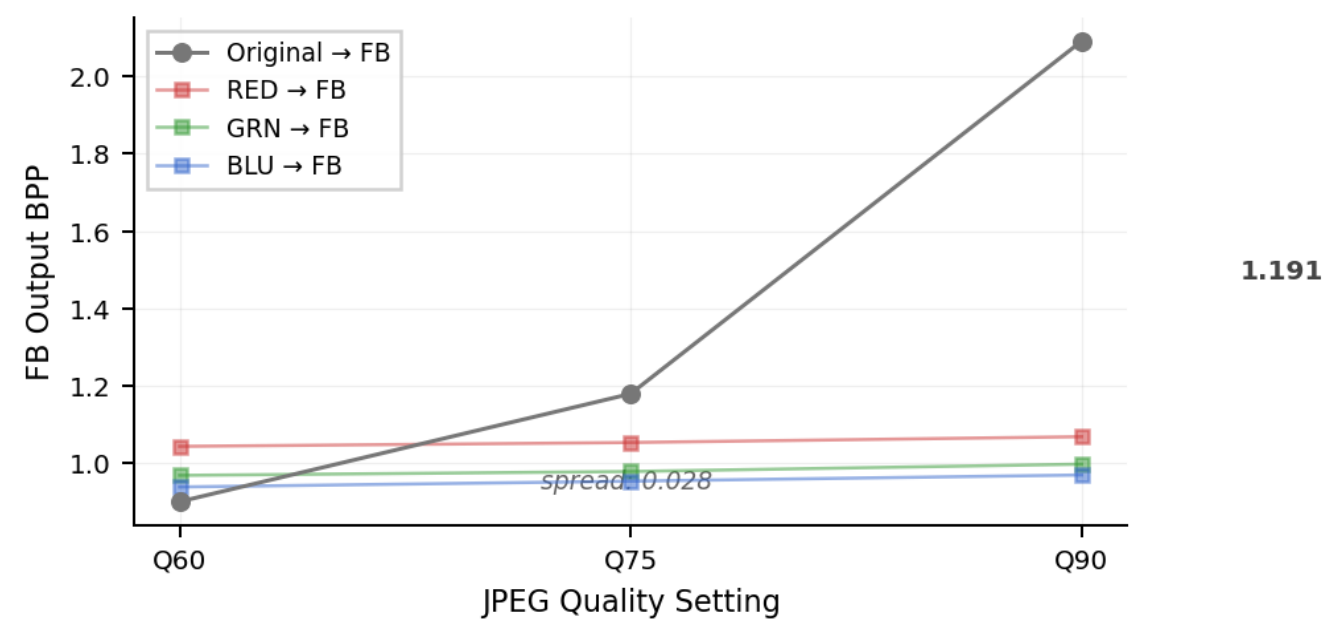
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	768x512	768x512	1.995	—
R-targeted → FB	2560x1707	2048x1366	1.068	46.5%
G-targeted → FB	2560x1707	2048x1366	0.997	50.0%
B-targeted → FB	2560x1707	2048x1366	0.969	51.4%
ABC (no perturbation) → FB	2560x1707	2048x1366	1.108	44.5%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.9169 vs 0.918 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.191 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.028 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim12

Best BPP (Q60, direct)	0.491	2560x1707, 4,369,920 px
Best BPP through FB	0.969	2048x1366 output
vs OG Q90 → FB	53.7%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216